The Moon · Staff Answers

Learning goal: I can explain how the Moon orbits the Earth and why we see different Moon phases.

Arrival Answers

1. The Earth.
2. About 28 days.
3. It reflects sunlight.
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. Half the Moon is still lit by the Sun, but that lit half faces the Sun while the dark side faces Earth.

Hinge: A. Normal phases happen because half of the Moon faces away from the Sun; we see different amounts of the lit half as it orbits Earth.

Evidence Desk Cards

Moon lies between the Sun and the Earth (Day 0): New Moon: we cannot see the lit side from Earth.

Slim curved sliver of light growing larger (Day 4): Waxing crescent: a small edge of the sunlit half is visible from Earth.

Moon is one quarter of the way round its orbit (Day 7): First quarter: we see half of the sunlit hemisphere (a half-circle in the sky).

Moon is on the opposite side of Earth from the Sun

(Day 14): Full Moon: we see the whole sunlit face as a bright circle.

More than half lit, shrinking after Full Moon (Day 18): Waning gibbous: most of the sunlit side is still visible, shrinking each night.

Phases happen because clouds or Earth's shadow cover the Moon every night: Misconception: normal phases are caused by our changing view of the Moon's sunlit half.

Check & Repair Answer

Earth's shadow points away from the Sun; a crescent Moon happens because we are looking mostly at the Moon's own unlit night side.

Exit Ticket Answers

1. About 28 days. 2. Sunlight reflected off the Moon's rocky surface. 3. When the Moon is opposite the Sun, the entire sunlit half of the Moon faces toward the Earth.